

Basic Information

Age Range
31-40 yearsSex
MaleWeight
85 kgHeight
164 cm

Procedure Information

Type of Procedure
ElectiveSurgical Branch
NeurosurgeryProcedure
Posterior fossa tumour excisionASA Grade
I

Incident Narrative Description

we had 40years male, no comorbidities, was posted for posterior fossa tumour excision. General Anaesthesia was given and procedure was uneventful. During the time of extubation, patient was slowly weaned. Neuromuscular recovery signs were confirmed, apart from the tidal volume, which kept being low for longer time. Simultaneously patient had respiratory distress and was desaturating. All the required measures for distress was taken, yet patient dint improve clinically. At this point, one of the anaesthetist team member remembered about the throat pack which was not removed. As soon as the throat pack was out, adequate tidal volume was achieved and saturation improved till 100%. Then patient was reversed and extubated.

Anesthesia Technique

Primary

- > General Anesthesia
- > ETT

Monitoring Used

ECG

Capnograph

Pulse Oximeter

IBP

Neuromuscular Monitor

Urine output

Incident Demographics

Time of Incident

- > Working hours

Place of Incident

- > Operating room

Timing of Event

- > During emergence

Incident Detection

- > Clinically by anesthesiologist

Incident Relation

- > To Anesthesia

Airway Incident

Airway Obstruction

Respiratory Related

Others: Respiratory distress

Anesthesia Procedure Related

Name of procedure: Throat packing

Describe incident: Forgot to remove the throat pack before extubation, which caused respiration distress and desaturation

Patient Outcome Category

Error, Harm

- > Category E - An error occurred that may have contributed to or resulted in temporary harm to the patient and required intervention.

Reviewer's Comments

Review #1

Incident & Outcome

Throat pack being forgotten prior to extubation.
Patient had episodes of desaturation

Contributing Factors

Forgetfulness and failure to inform the scrub nurse about the throat pack being placed.

Root Cause & Prevention

No objective way of reminding yourself about the presence of throat pack. Informing the scrub nurse after insertion of throat pack and making it a part of their count will enable better identification.

Suggested Course of Action

Not sure how and why the patient desaturated whilst the ETT was still insitu. The sequence of events would have correlated had the patient been extubated without removing the throat pack. So I'm assuming the sequence of events need to be clearer

Review #2

Incident & Outcome

Desaturation prior to reversal and extubation

Contributing Factors

Type of surgery - posterior cranial fossa tumor excision
Patient positioning - High chance of venous air embolism.
Drug : Muscle relaxant - duration of action/ timing of administration.
????? Throat pack

Root Cause & Prevention

DD:
1. Venous air embolism.(Supporting point : type of surgery, pt positioning; contradicting point : desaturation happened after closure)
2. incomplete reversal of neuromuscular blockade is likely .
3. Retension of throat pack. (Temporal description of event not matching)

Suggested Course of Action

For venous air embolism prevention and management :
Use of a pre-cordial doppler / TEE allows for early detection of intracardiac air.
Lowering the operating field below the level of the heart can help to prevent further entrainment of air.
A right atrial catheter can be used to therapeutically aspirate air from the right side of the heart; Increasing CVP with fluid administration may also be helpful in preventing further air entrainment by decreasing the pressure gradient.
Supportive therapy should be immediately implemented, including increasing the fraction of inspired oxygen to 100%.

For retention of surgical items the following actions can be in place :
The National Health Services, UK, have formulated a few guidelines for throat pack insertion which are mainly:
1. The indication for throat pack insertion should be discussed by an anesthesiologist and a surgeon, and it should have been documented with absolute indication
2. The individual making the decision assumes responsibility for the device. Trained anesthesiologist or surgeons are responsible for the insertion of the pack
3. There should be visual as well as documentary checks
4. As a visual check, the end of the pack should be left protruding from the mouth
5. The anesthesiologist documents insertion on the anesthetic chart
6. A "throat pack in situ" sticker is applied to the catheter mount
7. The anesthesiologist documents insertion on the swab board.

Reference

Taleska, G.; Trajkovska, T.; Nikolovska, A.; Shuplinoski, Z.; Micevski, M.; Sholjakova, M.; Pendovska, D.; Filipovski, I.; Strumenikovska, T.. Venous air embolism in patients undergoing posterior cranial fossa surgery in sitting position: A-365. European Journal of Anaesthesiology 23(0):p 96, June 2006.
Jagannathan S, Krovvidi H :Anaesthetic considerations for posterior fossa surgery
Continuing Education in Anaesthesia, Critical Care and Pain, 14, 202-206

Najjar MF, Kimpson J. A method for preventing throat pack retention. Anesth Analg. 1995 Jan;80(1):208-9. doi: 10.1097/00000539-199501000-00041. PMID: 7802289.
Crawford BS. Prevention of retained throat pack. Br Med J. 1977 Oct 15;2(6093):1029. doi: 10.1136/bmj.2.6093.1029. PMID: 922384; PMCID: PMC1631717.
Gray H, Brett C, Worthington J. Retained throat packs represent a potentially catastrophic airway hazard. Anaesth Intensive Care. 2006 Feb;34(1):119-20. PMID: 16494164.
Bhan, Sonia; Paul, Debashish; Singh, Shalendra; Gupta, Nipun. Throat Packs: Protocols Never to be Violated. Journal of Marine Medical Society 23(1):p 111-112, Jan-Jun 2021. | DOI: 10.4103/jmms.jmms_5_20